

Patch-seq Datasets Across Amygdala, Medial Temporal Lobe, and Brain Regions

Table of Contents

Introduction

Brain regions and cell populations relevant to amygdala and medial temporal lobe

Human medial temporal lobe and temporal cortex datasets

Cross-species and broader brain coverage

Data sources and availability

Introduction

Patch-seq links gene expression to how the same single cell fires and often how it looks. For a query on amygdala and medial temporal lobe cells, this matters because it helps identify which datasets go beyond transcriptomics and directly connect cell type with function. (4 sources)

Patch-seq is a single-cell method that combines patch-clamp electrophysiology with RNA sequencing from the same cell, rather than measuring only gene expression. It is used to connect transcriptomic identity with physiological properties, and in many studies also with cell morphology, which makes it especially useful for comparing neuron types across brain regions such as the amygdala, temporal cortex, and broader medial temporal lobe structures. ([Kobak et al., 2021](#)) ([Kobak et al., 2018](#)) ([Driessens et al., 2023](#)) ([Gamlin et al., 2025](#))

For your question, the key point is that relevant Patch-seq datasets are not limited to one data type: they can provide transcriptomic, electrophysiological, and sometimes morphological measurements for individual cells from human and other mammalian brain tissue. That makes them a strong fit when searching for datasets on cells from the amygdala or medial temporal lobe, because they can support both cell-type mapping and functional cell-state analysis in the same sampled neurons. ([Driessens et al., 2023](#)) ([Gamlin et al., 2025](#))

Evidence

(Kobak et al., 2021)

[Sparse reduced-rank regression for exploratory visualisation of paired multivariate data](#)

"One example is Patch-seq, a method combining single-cell RNA sequencing with electrophysiological recordings from the same cells."

(Kobak et al., 2018)

[Sparse reduced-rank regression for exploratory visualization of multimodal data sets](#)

"One example is Patch-seq, a method combining single-cell RNA sequencing with electrophysiological recordings from the same cells."

(Driessens et al., 2023)

[Genes associated with cognitive ability and HAR show overlapping expression patterns in human cortical neuron types](#)

"Patch-seq combines patch-clamp recording of single neurons with RNA sequencing of nuclear content of the same neuron and

thus allows not only access to transcriptomic information but also characterization of cellular physiological and morphological features (Fig. 2c)."

(Gamlin et al., 2025)

[Connectomics of predicted Sst transcriptomic types in mouse visual cortex](#)

"The Patch-seq technique enables the characterization of morphology, electrophysiology and transcriptomic properties from individual cells^{13, 14–15}."

Brain regions and cell populations relevant to amygdala and medial temporal lobe

Direct Patch-seq evidence in the provided papers covers several nearby or related regions and cell classes, but only one quote here names amygdala subregions explicitly. The strongest region-level leads for your query are medial/basomedial amygdala, hippocampal CA1, forebrain interneuron datasets, and newer temporal-lobe-heavy neuron profiling. (8 sources)

- **Amygdala subregions explicitly named:** the provided references directly mention the **medial amygdala (MEA)** and **basomedial amygdala (BMA)**, where **VGLUT2-class** and **GABA-class** neurons intermingle. This is the clearest amygdala-relevant signal in the set you gave. ([Hochgerner et al., 2023](#))
- **Medial temporal lobe relevance through hippocampus:** one Patch-seq analysis included a **novel dataset of 19 Pvalb-Cre positive interneurons from mouse hippocampal CA1**. Since CA1 is a core medial temporal lobe structure, this is directly relevant when you want Patch-seq cells from the broader MTL, even if it is not amygdala. ([Bomkamp et al., 2019](#))
- **Broader forebrain interneuron coverage:** Bomkamp et al. state that they used **four previously published Patch-seq datasets characterizing interneurons of the mouse forebrain**. That means useful Patch-seq cell types for an amygdala/MTL search may come from broader forebrain interneuron resources, not only datasets labeled with one exact structure name. ([Bomkamp et al., 2019](#))
- **Neocortical neuron classes also appear in early Patch-seq work:** one of the cited forebrain Patch-seq studies obtained full transcriptomes after patch clamp from **mouse neocortical pyramidal cells and interneurons**. This matters because temporal cortical parts of the medial temporal lobe search space may overlap more with these cortical Patch-seq resources than with amygdala-only datasets. ([Fuzik et al., 2015](#))
- **Interneuron-focused Patch-seq evidence extends to striatum:** another cited dataset used PatchSeq to study **Pvalb/Pthlh-related interneuron variation in striatum**, showing that Patch-seq datasets often center on inhibitory cell classes and can resolve subtype differences tied to physiology. This does not place cells in amygdala or MTL, but it helps set expectations for the kinds of cell populations commonly available. ([Bomkamp et al., 2019](#)) ([Munoz-Manchado et al., 2018](#))
- **GABAergic neurons are a common Patch-seq target class:** later Patch-seq resources highlighted in your reference set include **GABAergic mouse visual cortical neurons**, with RNA paired to electrophysiology and, in some datasets, local morphology. This again is not amygdala-specific, but it shows a strong bias toward inhibitory neuron profiling that may carry over to temporal-lobe or amygdala-oriented searches. ([Gliko et al., 2024](#)) ([Hu et al., 2024](#)) ([Gouwens et al., 2020](#))
- **Temporal-lobe-heavy profiling is now present in newer datasets:** one recent paper states, "**We profiled 1491 neurons, primarily in temporal lobe.**" This is the most direct temporal-lobe region statement in your reference set and suggests that if your goal is medial temporal lobe-adjacent Patch-seq cells, newer temporal-lobe-focused datasets may be especially useful. ([Dalley et al., 2026](#))
- **Practical takeaway for your query:** based on these references, the most relevant cell populations to look for are **GABAergic interneurons, Pvalb-class cells, and mixed excitatory/inhibitory amygdala populations**, with region labels that may include **MEA, BMA, CA1 hippocampus, temporal lobe, and forebrain/temporal**

cortex rather than only the exact phrase "medial temporal lobe amygdala." ([Hochgerner et al., 2023](#)) ([Bomkamp et al., 2019](#)) ([Dalley et al., 2026](#))

Evidence

([Hochgerner et al., 2023](#))

[Neuronal types in the mouse amygdala and their transcriptional response to fear conditioning](#)

"In the medial amygdala (MEA) and basomedial amygdala (BMA), neurons from VGLUT2 and GABA classes intermixed."

([Bomkamp et al., 2019](#))

[Transcriptomic correlates of electrophysiological and morphological diversity within and across excitatory and inhibitory neuron classes](#)

"Here, we made use of four previously published PatchSeq datasets that have characterized interneurons of the mouse forebrain, described in detail in Table 1. ("Cadwell," "Földy," "Fuzik," "Muñoz"; (Cadwell et al., 2015)(Fuzik et al., 2015)(Földy et al., 2016) (Muñoz-Manchado et al., 2018)). Our analysis also includes one novel dataset of 19 Pvalb-Cre positive interneurons recorded in region CA1 of the mouse hippocampus, reported here for the first time."

([Fuzik et al., 2015](#))

[Integration of electrophysiological recordings with single-cell RNA-seq data identifies novel neuronal subtypes](#)

"Traditionally, neuroscientists have defined the identity of neurons by the cells' location, morphology, connectivity and excitability. However, the direct relationship between these parameters and the molecular phenotypes has remained largely unexplored. Here, we present a method for obtaining full transcriptome data from single neocortical pyramidal cells and interneurons after whole-cell patch-clamp recordings in mouse brain slices. In our approach, termed Patch-seq, a patch-clamp stimulus protocol is followed by the aspiration of the entire somatic compartment into the recording pipette, reverse transcription of RNA including addition of unique molecular identifiers, cDNA amplification, Illumina library preparation and sequencing. We show that Patch-seq reveals a close link between electrophysiological characteristics, responses to acute chemical challenges and RNA expression of neurotransmitter receptors and channels. Moreover, it distinguishes neuronal subpopulations that correspond to both well-established and, to our knowledge, hitherto undescribed neuronal subtypes. Our findings demonstrate the ability of Patch-seq to precisely map neuronal subtypes and predict their network contributions in the brain."

([Munoz-Manchado et al., 2018](#))

[Diversity of Interneurons in the Dorsal Striatum Revealed by Single-Cell RNA Sequencing and PatchSeq](#)

"Summary Striatal locally projecting neurons, or interneurons, act on nearby circuits and shape functional output to the rest of the basal ganglia. We performed single-cell RNA sequencing of striatal cells enriching for interneurons. We find seven discrete interneuron types, six of which are GABAergic. In addition to providing specific markers for the populations previously described, including those expressing Sst/Npy, Th, Npy without Sst, and Chat, we identify two small populations of cells expressing Cck with or without Vip. Surprisingly, the Pvalb-expressing cells do not constitute a discrete cluster but rather are part of a larger group of cells expressing Pthlh with a spatial gradient of Pvalb expression. Using PatchSeq, we show that Pthlh cells exhibit a continuum of electrophysiological properties correlated with expression of Pvalb. Furthermore, we find significant molecular differences that correlate with differences in electrophysiological properties between Pvalb-expressing cells of the striatum and those of the cortex."

([Gliko et al., 2024](#))

[High-throughput analysis of dendrite and axonal arbors reveals transcriptomic correlates of neuroanatomy](#)

"The dataset profiling local morphology and transcriptome of GABAergic mouse visual cortical neurons was generated as part of the Patch-seq recordings described in ref. 5."

([Hu et al., 2024](#))

[scPair: Boosting single cell multimodal analysis by leveraging implicit feature selection and single cell atlases](#)

"Gouwens et al. (Gouwens et al., 2020) collected Patch-seq profiles consisting of RNA paired with electrophysiological recordings for GABAergic neurons from the mouse primary visual cortex."

([Gouwens et al., 2020](#))

[Integrated Morphoelectric and Transcriptomic Classification of Cortical GABAergic Cells](#)

"Neurons are frequently classified into distinct types on the basis of structural, physiological, or genetic attributes. To better constrain the definition of neuronal cell types, we characterized the transcriptomes and intrinsic physiological properties of over

4,200 mouse visual cortical GABAergic interneurons and reconstructed the local morphologies of 517 of those neurons. We find that most transcriptomic types (t-types) occupy specific laminar positions within visual cortex, and, for most types, the cells mapping to a t-type exhibit consistent electrophysiological and morphological properties. These properties display both discrete and continuous variation among t-types. Through multimodal integrated analysis we define 28 met-types that have congruent morphological, electrophysiological, and transcriptomic properties and robust mutual predictability. We identify layer-specific axon innervation pattern as a defining feature distinguishing different met-types. These met-types represent a unified definition of cortical GABAergic interneuron types, providing a systematic framework to capture existing knowledge and bridge future analyses across different modalities."

(Dalley et al., 2026)

[Human Neocortical Glutamatergic Neurons Revealed Through Multimodal Profiling](#)

"We profiled 1491 neurons, primarily in temporal lobe"

Human medial temporal lobe and temporal cortex datasets

The clearest human Patch-seq evidence in your references comes from neurosurgically resected temporal cortical tissue, especially middle temporal gyrus datasets tied to a human transcriptomic taxonomy. These studies are highly relevant for medial-temporal-lobe-adjacent searches, but they are strongest for temporal cortex and MTG rather than explicitly labeled amygdala samples. (4 sources)

For human data, the most direct Patch-seq platform in your reference set is the study by Berg et al., which built a workflow that combines patch-clamp recording, biocytin staining, and single-cell RNA-seq on **neurosurgically resected human tissue**. That means the resulting cells can link physiology, morphology, and transcriptomic identity in the same human neurons, which is exactly the kind of dataset many people mean when they ask for Patch-seq cells from human temporal lobe tissue. ([Berg et al., 2021](#))

A key temporal-cortex dataset highlighted in the references is the human **middle temporal gyrus (MTG)** Patch-seq resource used by Arbabi et al. They note that human Patch-seq transcriptomes were mapped to a reference taxonomy based on **single-nucleus RNA-seq from the human medial temporal gyrus**, and that their analysis included **278 supragranular glutamatergic neurons from MTG across 48 subjects**. This is one of the clearest human temporal-lobe Patch-seq datasets in the material you provided. ([Arbabi et al., 2022](#)) ([Berg et al., 2021](#)) ([Hodge et al., 2019](#))

That MTG mapping step also matters for dataset discovery: even when a Patch-seq paper is focused on physiology or morphology, the cell labels may be assigned through the **Hodge et al. human MTG taxonomy**, which is a notable reference atlas for human temporal cortex cell types. So for a search on medial temporal lobe or nearby temporal-lobe brain regions, datasets may be indexed under **MTG, medial temporal gyrus**, or under taxonomy labels inherited from that human cortex reference rather than under the exact disease or surgery label. ([Arbabi et al., 2022](#)) ([Hodge et al., 2019](#))

A newer data-integration paper makes this even more practical: Miller et al. describe a "**Middle temporal gyrus (MTG) (initial studies)**" table that links taxonomies across two MTG studies and notes that the earlier MTG taxonomy has been used as a reference for **several human Patch-seq studies**. They also state that annotation tables from two Patch-seq studies are included, which suggests that human temporal-cortex Patch-seq datasets can often be connected across releases through shared MTG taxonomies and label-transfer tools rather than treated as isolated datasets. ([Miller et al., 2025](#)) ([Hodge et al., 2019](#))

Taken together, the main human lead from your references is not a dedicated human amygdala Patch-seq atlas, but a **strong temporal cortex / MTG Patch-seq ecosystem** based on surgical human tissue, multimodal single-cell measurements, and transcriptomic mapping to human temporal gyrus reference atlases. For your query, that means the best-supported human datasets are currently those from **human temporal cortex, especially MTG and medial temporal gyrus-linked taxonomies**, which can still be very useful when you want medial-temporal-lobe-adjacent human neuron data with electrophysiology and morphology attached. ([Berg et al., 2021](#)) ([Arbabi et al., 2022](#)) ([Miller et al., 2025](#))

Evidence

(Berg et al., 2021)

[Human neocortical expansion involves glutamatergic neuron diversification](#)

"we developed a robust platform combining patch clamp recording, biocytin staining and single-cell RNA-sequencing (Patch-seq) to examine neurosurgically resected human tissues."

(Arbabi et al., 2022)

[Investigating microglia-neuron crosstalk by characterizing microglial contamination in human and mouse patch-seq datasets](#)

"For the human Patch-seq data from Allen Institute for Brain Science"

"Single-cell transcriptomes from Patch-seq samples were subsequently mapped to a reference taxonomy based on single-nucleus RNA-seq data from the human medial temporal gyrus. (Berg et al., 2021)(Hodge et al., 2019) RNA-sequencing metadata and counts data were downloaded for our analyses (accessed 05/10/ 2021). A total of 278 supragranular glutamatergic neurons from MTG in N = 48 subjects (37.1 G 14.3 years, 18 female, 16 Caucasian, 1 American Indian and Alaska Native, 31 Unknown or unreported) were included."

(Hodge et al., 2019)

[Conserved cell types with divergent features in human versus mouse cortex](#)

"Elucidating the cellular architecture of the human cerebral cortex is central to understanding our cognitive abilities and susceptibility to disease. Here we used single-nucleus RNA-sequencing analysis to perform a comprehensive study of cell types in the middle temporal gyrus of human cortex. We identified a highly diverse set of excitatory and inhibitory neuron types that are mostly sparse, with excitatory types being less layer-restricted than expected. Comparison to similar mouse cortex single-cell RNA-sequencing datasets revealed a surprisingly well-conserved cellular architecture that enables matching of homologous types and predictions of properties of human cell types. Despite this general conservation, we also found extensive differences between homologous human and mouse cell types, including marked alterations in proportions, laminar distributions, gene expression and morphology. These species-specific features emphasize the importance of directly studying human brain. RNA-sequencing analysis of cells in the human cortex enabled identification of diverse cell types, revealing well-conserved architecture and homologous cell types as well as extensive differences when compared with datasets covering the analogous region of the mouse brain."

(Miller et al., 2025)

[Annotation Comparison Explorer \(ACE\): connecting brain cell types across studies of health and Alzheimer's Disease](#)

"The "Middle temporal gyrus (MTG) (initial studies)" table connected taxonomies from two distinct studies of MTG by using MapMyCells (RRID:SCR_024672; <https://portal.brainmap.org/atlas-and-data/bkp/mapmycells>) to transfer labels from the recent human MTG SEA-AD taxonomy (Gabbitto et al., 2024) to cells in an earlier MTG taxonomy (Hodge et al., 2019) that has been used as a reference for several human patch-seq studies"

". Annotation tables from two patch-seq studies are included as well (Gouwens et al., 2020)(Scala et al., 2020)."

Cross-species and broader brain coverage

Patch-seq is not limited to one species or one cortical area. For an amygdala/medial temporal lobe search, the main value of these papers is that they show the method and public resources extend across mouse, human, macaque, and other primates, so nearby-region and cross-species comparison datasets can still be very useful. (12 sources)

- **Patch-seq is broadly portable across species and regions.** One protocol paper states directly that the method is "generalizable to different species and brain regions" and demonstrates multimodal capture from **human and macaque brain slices**. This is important for your query because even if amygdala-specific Patch-seq data are sparse, the method itself is not restricted to one taxon or one cortical area. ([Lee et al., 2021](#))
- **Large public Patch-seq resources are strongest in mouse cortex.** A 2022 analysis says it used two of the largest public Patch-seq datasets, one from **mouse motor cortex** and one from **mouse visual cortex**. Those datasets give broad reference coverage for multimodal neuron typing, even when they are not from amygdala or medial temporal lobe. ([Lee et al., 2022](#)) ([Scala et al., 2020](#)) ([Gouwens et al., 2020](#))
- **The mouse visual cortex dataset is especially notable in scale.** Gouwens et al. profiled over 4,200 mouse

visual cortical GABAergic interneurons and reconstructed local morphology for 517 of them, defining integrated morpho-electric-transcriptomic cell groups. That makes it a strong broader-brain reference for inhibitory neuron phenotypes that may help interpret interneuron classes found in temporal lobe or amygdala-adjacent tissue. ([Gouwens et al., 2020](#))

- **Mouse motor cortex provides another large multimodal reference atlas.** Scala et al. used Patch-seq on over 1,300 neurons in adult mouse motor cortex and linked transcriptomic types to morphology and physiology across nearly all major cortical classes. This is useful as a comparison set when a target region dataset is smaller or less complete. ([Scala et al., 2020](#))
- **Cross-species work now includes human, mouse, marmoset, and macaque lineages.** Reviews and overview papers describe Patch-seq use in rodents, adult human, and non-human primate brain slices, and broader BICCN work used cross-species cell type classification in humans, marmosets, and mice while using Patch-seq to identify specialized primate and human motor-cortex neurons. ([Shen et al., 2024](#)) ([Bakken et al., 2021](#))
- **Human-to-rodent comparison is an active use case, not just a technical possibility.** Human Patch-seq studies of cortical neurons report both conserved and divergent features relative to rodent homologs, including for layer 5 extratelencephalic-projecting neurons and broad GABAergic neuron types. So if your main region of interest is human amygdala or MTL, nearby human temporal datasets can still be compared against mouse resources in a principled way. ([Kalmbach et al., 2021](#)) ([Lee et al., 2023](#))
- **Human Patch-seq coverage now extends beyond pyramidal neurons.** Summary papers note large-scale success for human pyramidal neurons, and newer work also profiled hundreds of human cortical GABAergic neurons with transcriptome, physiology, and morphology from the same cells. This matters because amygdala and medial temporal lobe studies often care about both excitatory and inhibitory populations. ([Camunas-Soler, 2023](#)) ([Berg et al., 2021](#)) ([Kalmbach et al., 2021](#)) ([Lee et al., 2023](#))
- **Non-human primate coverage is growing beyond macaque.** A recent study reports Patch-seq recordings from supragranular pyramidal neurons in one New-World monkey species and two Old-World monkey species, showing that primate comparative datasets are expanding. ([Radaelli et al., 2025](#))
- **Broader coverage also includes in vitro human systems.** One benchmarking paper notes that some early Patch-seq datasets came from human stem-cell-derived neurons and astrocytes in culture, alongside mouse acute-slice datasets. These are not region-resolved brain samples, but they can still help for method development or gene-function comparisons. ([Tripathy et al., 2018](#))
- **Practical takeaway for your query:** if you do not find many Patch-seq cells explicitly labeled amygdala or medial temporal lobe, it is still worth checking human temporal cortex, mouse cortex, and primate comparative Patch-seq resources. The papers here show that Patch-seq supports cross-species mapping and broad brain coverage, so useful reference cells may come from outside the exact target label while still helping classify or interpret amygdala/MTL neurons. ([Lee et al., 2021](#)) ([Shen et al., 2024](#)) ([Camunas-Soler, 2023](#))

Evidence

(Lee et al., 2021)

[Scaled, high fidelity electrophysiological, morphological, and transcriptomic cell characterization](#)

"The protocol is generalizable to different species and brain regions, as demonstrated by capturing multimodal data from human and macaque brain slices."

(Lee et al., 2022)

[MorphNet Predicts Cell Morphology from Single-Cell Gene Expression](#)

"Patch-seq is an experimental method that can simultaneously profile single neuron gene expression, electrophysiology, and morphology (Fuzik et al., 2015)(Lipovsek et al., 2021)"

"In our work, we analyzed two of the largest publicly available Patch-seq datasets, one from the mouse motor cortex (Scala et al., 2020) and other from the mouse visual cortex (Gouwens et al., 2020) ."

(Scala et al., 2020)

Phenotypic variation within and across transcriptomic cell types in mouse motor cortex

"Cortical neurons exhibit astounding diversity in gene expression as well as in morphological and electrophysiological properties. Most existing neural taxonomies are based on either transcriptomic or morpho-electric criteria, as it has been technically challenging to study both aspects of neuronal diversity in the same set of cells. Here we used Patch-seq to combine patch-clamp recording, biocytin staining, and single-cell RNA sequencing of over 1300 neurons in adult mouse motor cortex, providing a comprehensive morpho-electric annotation of almost all transcriptomically defined neural cell types. We found that, although broad families of transcriptomic types (Vip, Pvalb, Sst, etc.) had distinct and essentially non-overlapping morpho-electric phenotypes, individual transcriptomic types within the same family were not well-separated in the morpho-electric space. Instead, there was a continuum of variability in morphology and electrophysiology, with neighbouring transcriptomic cell types showing similar morpho-electric features, often without clear boundaries between them. Our results suggest that neural types in the neocortex do not always form discrete entities. Instead, neurons follow a hierarchy consisting of distinct non-overlapping branches at the level of families, but can form continuous and correlated transcriptomic and morpho-electrical landscapes within families."

(Gouwens et al., 2020)

Integrated Morphoelectric and Transcriptomic Classification of Cortical GABAergic Cells

"Neurons are frequently classified into distinct types on the basis of structural, physiological, or genetic attributes. To better constrain the definition of neuronal cell types, we characterized the transcriptomes and intrinsic physiological properties of over 4,200 mouse visual cortical GABAergic interneurons and reconstructed the local morphologies of 517 of those neurons. We find that most transcriptomic types (t-types) occupy specific laminar positions within visual cortex, and, for most types, the cells mapping to a t-type exhibit consistent electrophysiological and morphological properties. These properties display both discrete and continuous variation among t-types. Through multimodal integrated analysis we define 28 met-types that have congruent morphological, electrophysiological, and transcriptomic properties and robust mutual predictability. We identify layer-specific axon innervation pattern as a defining feature distinguishing different met-types. These met-types represent a unified definition of cortical GABAergic interneuron types, providing a systematic framework to capture existing knowledge and bridge future analyses across different modalities."

(Shen et al., 2024)

Multimodal Nature of the Single-cell Primate Brain Atlas: Morphology, Transcriptome, Electrophysiology, and Connectivity

"Recently, several groups have developed and optimized Patch-seq, a multimodal method that describes electrophysiological, transcriptomic, and morphological profiles in single neurons of rodents (Cadwell et al., 2015)(Cadwell et al., 2017)(Fuzik et al., 2015) [59](Lipovsek et al., 2020) and adult human and non-human primate brain slices (Berg et al., 2021)59,(Chartrand et al., 2022)(Lee et al., 2023)(Markram et al., 2004)(Kalmbach et al., 2021)(Hurk et al., 2018)."

(Bakken et al., 2021)

Comparative cellular analysis of motor cortex in human, marmoset and mouse

"The primary motor cortex (M1) is essential for voluntary fine-motor control and is functionally conserved across mammals¹. Here, using high-throughput transcriptomic and epigenomic profiling of more than 450,000 single nuclei in humans, marmoset monkeys and mice, we demonstrate a broadly conserved cellular makeup of this region, with similarities that mirror evolutionary distance and are consistent between the transcriptome and epigenome. The core conserved molecular identities of neuronal and non-neuronal cell types allow us to generate a cross-species consensus classification of cell types, and to infer conserved properties of cell types across species. Despite the overall conservation, however, many species-dependent specializations are apparent, including differences in cell-type proportions, gene expression, DNA methylation and chromatin state. Few cell-type marker genes are conserved across species, revealing a short list of candidate genes and regulatory mechanisms that are responsible for conserved features of homologous cell types, such as the GABAergic chandelier cells. This consensus transcriptomic classification allows us to use patch-seq (a combination of whole-cell patch-clamp recordings, RNA sequencing and morphological characterization) to identify corticospinal Betz cells from layer 5 in non-human primates and humans, and to characterize their highly specialized physiology and anatomy. These findings highlight the robust molecular underpinnings of cell-type diversity in M1 across mammals, and point to the genes and regulatory pathways responsible for the functional identity of cell types and their species-specific adaptations. An examination of motor cortex in humans, marmosets and mice reveals a generally conserved cellular makeup that is likely to extend to many mammalian species, but also differences in gene expression, DNA methylation and chromatin state that lead to species-dependent specializations."

(Kalmbach et al., 2021)

Signature morpho-electric, transcriptomic, and dendritic properties of human layer 5 neocortical pyramidal neurons

"Summary In the neocortex, subcerebral axonal projections originate largely from layer 5 (L5) extratelencephalic-projecting (ET) neurons. The unique morpho-electric properties of these neurons have mainly been described in rodents, where retrograde tracers or transgenic lines can label them. Similar labeling strategies are infeasible in the human neocortex, rendering the translational relevance of findings in rodents unclear. We leveraged the recent discovery of a transcriptomically-defined L5 ET neuron type to study the properties of human L5 ET neurons in neocortical brain slices derived from neurosurgeries. Patch-seq recordings, where transcriptome, physiology and morphology were assayed from the same cell, revealed many conserved morpho-electric properties of human and rodent L5 ET neurons. Divergent properties were often subtler than differences between L5 cell types within these two species. These data suggest a conserved function of L5 ET neurons in the neocortical hierarchy, but also highlight phenotypic divergence possibly related to functional specialization of human neocortex."

(Lee et al., 2023)

Signature morphoelectric properties of diverse GABAergic interneurons in the human neocortex

"Human cortex transcriptomic studies have revealed a hierarchical organization of γ -aminobutyric acid–producing (GABAergic) neurons from subclasses to a high diversity of more granular types. Rapid GABAergic neuron viral genetic labeling plus Patch-seq (patch-clamp electrophysiology plus single-cell RNA sequencing) sampling in human brain slices was used to reliably target and analyze GABAergic neuron subclasses and individual transcriptomic types. This characterization elucidated transitions between PVALB and SST subclasses, revealed morphological heterogeneity within an abundant transcriptomic type, identified multiple spatially distinct types of the primate-specialized double bouquet cells (DBC), and shed light on cellular differences between homologous mouse and human neocortical GABAergic neuron types. These results highlight the importance of multimodal phenotypic characterization for refinement of emerging transcriptomic cell type taxonomies and for understanding conserved and specialized cellular properties of human brain cell types. Description INTRODUCTION Recent studies using single-cell transcriptomic analysis have shed new light on genetically defined cell types in the human brain. However, a deeper understanding of multimodal cellular properties, such as electrical activity and morphology, remains elusive and is central to understanding the role of distinct cell types in cognitive function and disease. RATIONALE Understanding the fundamental properties of different cell types is important for gaining insights into their role in neural circuits, cognition, and disease. This is challenging for cell types in the human brain because of the limited availability of tissue, the heterogeneity of neurons, and the lack of genetic approaches to effectively target these spatially distributed and in some cases, exceedingly rare types. Rapid viral genetic labeling and patch-clamp electrophysiology combined with RNA sequencing (Patch-seq) facilitates the targeting of specific cell types in tissue from human neurosurgeries and thus allows the characterization of multimodal properties of neurons in the human cortex. RESULTS Patch-seq sampling facilitated targeted acquisition and analysis of 778 human neurons in cortical layers 2 to 6, across 44 out of 45 γ -aminobutyric acid–producing (GABAergic) transcriptomic types. Aggregated data from acute and culture paradigms provide new descriptions and direct comparison of the signature morphoelectric properties of the canonical human interneuron subclasses, LAMP5/PAX6, VIP, SST, and PVALB, and a deeper dive into features of select GABAergic transcriptomic types. Detailed characterization of the SST subclass revealed specific multimodal properties of individual transcriptomic types. The SST FRZB transcriptomic type exhibits gene expression signatures along a continuum of PVALB and SST subclasses, but the morphoelectric properties of this type clearly indicate strong alignment with the PVALB subclass. For an additional transcriptomic type, SST CALB1, we found multiple discrete morphological types suggesting that further splitting may be warranted. Double-bouquet cells of the human neocortex belong to two transcriptomic types within the SST subclass with enrichment in temporal versus frontal cortex regions. We compared the morphoelectric properties of homologous mouse and human neocortical GABAergic neuron types and found that human types are more excitable and have a larger spatial extent with less neurite branching. CONCLUSION An impactful finding of our study is the direct demonstration of how multimodal Patch-seq data is vital to refinement of transcriptomic cell–type taxonomies. Cellular taxonomies built on single-cell transcriptomes and differential gene expression are not static but rather represent a starting foundation to build upon as new data modalities are obtained and aligned at the resolution of transcriptomic types. We also demonstrate the immense potential and utility of viral genetic labeling and Patch-seq for targeted analysis of human neocortical GABAergic neuron subclasses and types in ex vivo brain slices. This work can serve as a roadmap for future functional studies of human brain cell types at the resolution of emerging transcriptomic cell–type taxonomies and provides a rich open-access dataset for exploring gene-function relationships for a wide diversity of human neocortical GABAergic neuron types. Patch-seq analysis of human GABAergic neurons. (A) MRIs are obtained to accurately map the area of resected human brain tissue. (B) Viral labeling combined with Patch-seq is used to collect the multimodal properties of human cortical GABAergic neurons. Each sample is mapped to a taxonomy to identify the transcriptomic type. (C) Representative GABAergic subclasses, their morphologies, and action-potential firing patterns. (D) Transcriptomic type SST FRZB is found along a continuum of electrophysiological and morphological features between the SST and PVALB subclasses. (E) Morphologically defined double-bouquet cells map to both SST CALB1 and SST ADGRG6, each displaying a distinct distribution across the cortical layers. (F) Homology mapping of PVALB types between mice and humans reveals morphological and electrophysiological differences, including larger spatial extent and higher excitability of human neurons."

(Camunas-Soler, 2023)

Integrating single-cell transcriptomics with cellular phenotypes: cell morphology, Ca²⁺ imaging and electrophysiology

"These large-scale initiatives have succeeded at obtaining patchseq data for thousands of neurons in the primary motor and visual cortex in rodents (Gouwens et al., 2020) BRAIN Initiative Cell Census Network (BICCN) 2021; (Bakken et al., 2021) as well as

human pyramidal neurons (Berg et al., 2021)(Kalmbach et al., 2021)."

(Berg et al., 2021)

[Human neocortical expansion involves glutamatergic neuron diversification](#)

"we developed a robust platform combining patch clamp recording, biocytin staining and single-cell RNA-sequencing (Patch-seq) to examine neurosurgically resected human tissues."

(Radaelli et al., 2025)

[HCN channels reveal conserved and divergent physiology in supragranular pyramidal neurons in primate species](#)

"Using Patch-seq recordings from acute and cultured brain slices, we found robust HCN-dependent physiological properties in supragranular pyramidal neurons in a species of New-World monkey (*Saimiri sciureus*) and two species of Old-World Monkey (*Macaca mulatta*, *Macaca nemestrina*)."

(Tripathy et al., 2018)

[Assessing Transcriptome Quality in Patch-Seq Datasets](#)

"The Cadwell, Földy, and Fuzik datasets were collected from acute brain slices from adult and juvenile mice whereas the Bardy and Chen datasets were collected from human stem-cell derived neurons and astrocytes in culture."

Data sources and availability

The clearest data-access lead in the provided references is a public GitHub repository from the Allen Institute. For your amygdala or medial temporal lobe search, this means at least some Patch-seq resources are distributed as reusable public data/code packages rather than only as figures in papers. (1 source)

- **Public access point:** Covert et al. state that **"The Patchseq data is available at** This gives you a concrete starting place for downloading or inspecting Patch-seq data resources tied to their analysis. ([Covert et al., 2022](#))
- **How to use this for your query:** because your target is **amygdala / medial temporal lobe brain cells**, a practical next step is to use that repository as a reference point for data format, metadata fields, and linked Patch-seq resources, then check whether region labels in the included or related datasets match terms such as **temporal lobe, MTG, hippocampus, or amygdala**. ([Covert et al., 2022](#))
- **Practical takeaway:** in the papers you provided, the strongest explicit availability statement is for an **Allen Institute Patch-seq GitHub resource**. So for finding relevant cells, it makes sense to search not only paper titles and abstracts, but also linked repositories and supplementary metadata where region annotations are often easier to filter. ([Covert et al., 2022](#))

Evidence

(Covert et al., 2022)

[Predictive and robust gene selection for spatial transcriptomics](#)

"The Patchseq data is available at <https://github.com/AllenInstitute/coupledAE-patchseq>."